

Input Handling & Data Acquisition

Possible input:

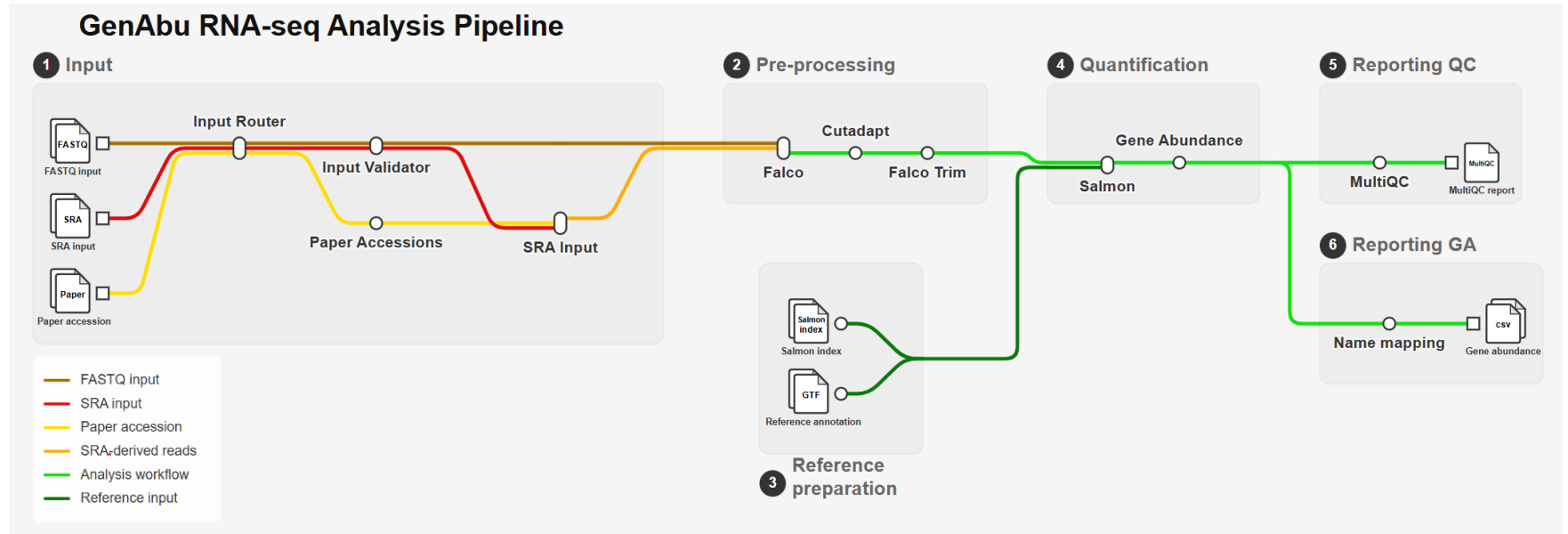
- FastQ
- SRA accessions
- research paper

Paper accession:

- pdf to text converter
- SRA accession

Next steps:

- samples from different species in one run
- input validation for samplesheet of paper input



ffq:

- source accession (e.g. GSE223541) → structured metadata
- metadata: run accession, scientific name (species), library strategy (RNA seq), library layout, ftp path

nf-test

- automated end-to-end testing of the complete pipeline, unit testing and integration testing
- setup:
 - defining test objects
 - selecting test datasets
 - creating test cases
 - executing pipeline with test cases
- Continuous Integration (CI)
- stub mode verifies the process contract without running the modules

```
test("Should reject more than two FASTQ files") {  
  when {  
    process {  
      ""  
      input[0] = Channel.of(tuple(  
        [  
          id: 'invalid',  
          single_end: false  
        ],  
        [  
          file(  
            '${pairedRead1.canonicalPath}',  
            checkIfExists: true  
          ),  
          file(  
            '${pairedRead2.canonicalPath}',  
            checkIfExists: true  
          ),  
          file(  
            '${thirdRead.canonicalPath}',  
            checkIfExists: true  
          )  
        ]  
      )  
    }  
  }  
  then {  
    assert process.failed  
  }  
}
```